

BOX OFFICE ANALYSIS FOR STRATEGIC MOVIE PRODUCTION

- Using historical movie performance to guide profitable production decisions

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OVERVIEW

Purpose of This Analysis:

- Support data-driven production decisions
- Identify drivers of box office success
- Reduce financial risk

What We Delivered:

- Analysis of historical movie data
- Actionable business recommendations

PROJECT OBJECTIVES

Goal 1:

Identify movie genres that generate the highest worldwide revenue to guide prioritization.

Goal 2:

Analyze the correlation between IMDb ratings and box office revenue to understand how audience approval impacts financial success.

Goal 3:

Determine the optimal movie runtime range for commercial performance.

BUSINESS PROBLEM

The Challenge:

- Movie production is capital-intensive and high-risk
- Poor genre, quality, or runtime choices lead to losses

Key Questions:

- Which genres generate the highest revenue?
- Do ratings influence revenue?
- What runtime performs best?

BUSINESS UNDERSTANDING

A new movie studio wants to make data-driven decisions to maximize box office success.

We analyzed historical movie data to uncover patterns in:

1. Genre popularity and revenue potential
2. The impact of audience ratings on financial performance
3. Ideal movie runtimes for commercial appeal

DATA UNDERSTANDING

Data Sources:

- IMDB SQLite database (im.db)
- Box Office Mojo dataset (bom.movie_gross.csv.gz)

Cleaning & Merging:

1. Filtered IMDB data to include only movie-like entries
2. Cleaned and standardized titles for merging
3. Combined datasets with a 29.41% match rate
4. Created worldwide_gross from domestic and foreign revenue

DATA ANALYSIS I: GENRE PERFORMANCE

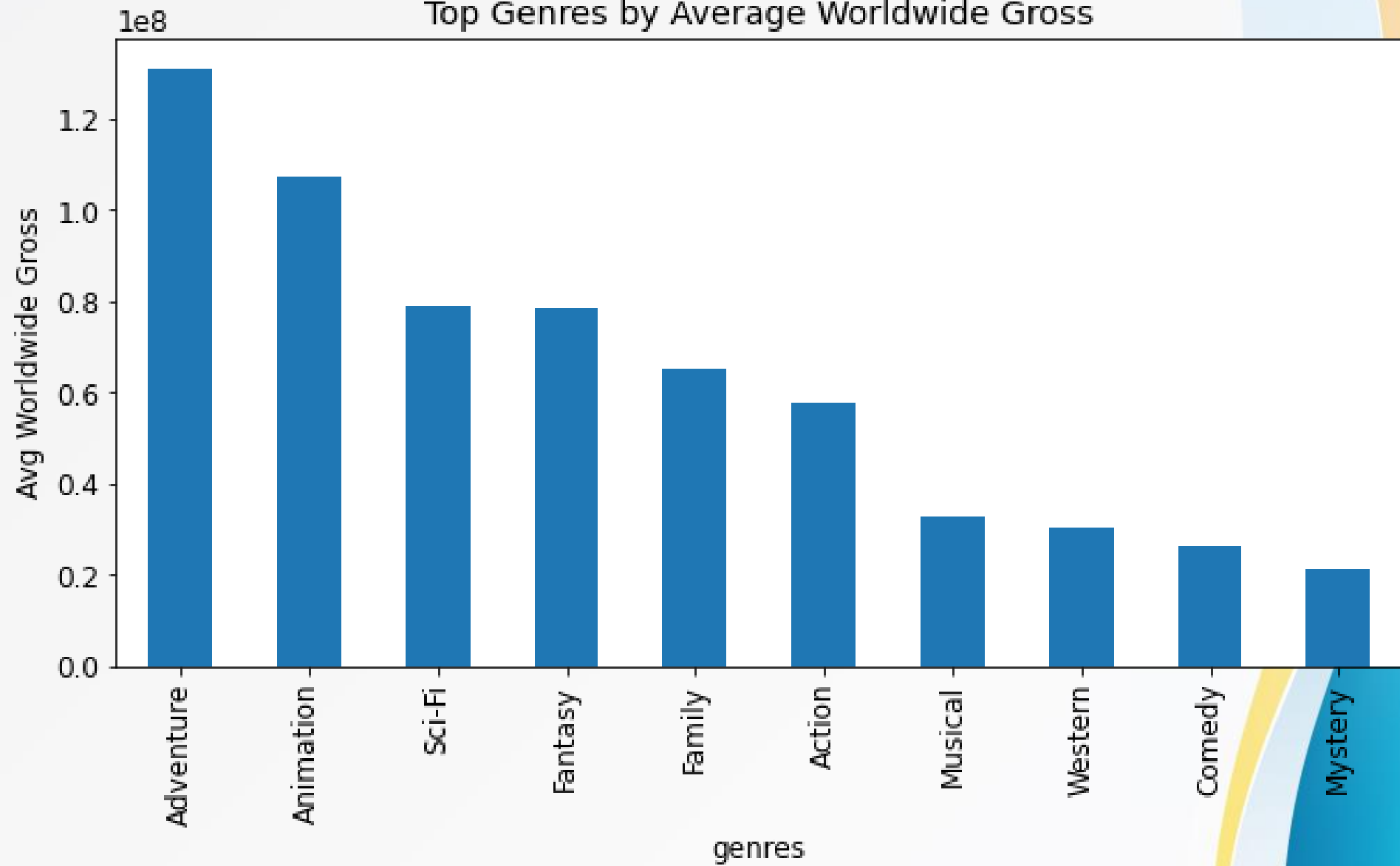
Findings:

- Specific genres dominate the top-performing films globally and deliver significantly higher financial returns than others.
- Adventure, Animation, and Sci-Fi earn the highest revenue
- These genres dominate global box office results

Recommendation:

- The studio should focus on these high-earning genres to increase the likelihood of competing effectively with major established studios.

Top Genres by Average Worldwide Gross



DATA ANALYSIS II: RATINGS VS REVENUE

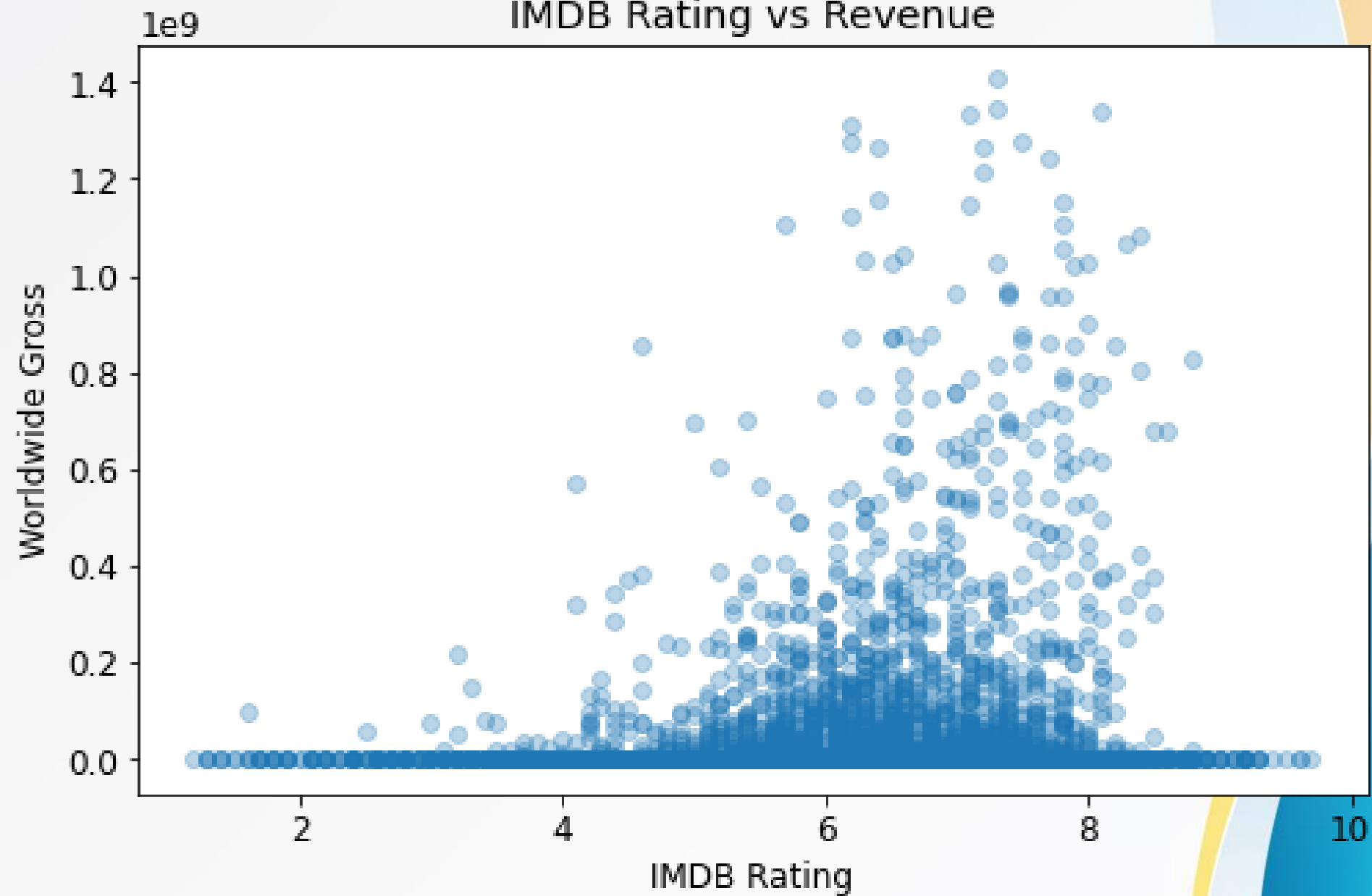
Findings:

- Movies with IMDb ratings in the 6.5–8.0 range produce the strongest and most consistent box-office returns.
- Very low ratings underperform
- High ratings alone do not guarantee success

Recommendation:

- Focus on quality storytelling and strong direction to achieve ratings above 7.5, which drive higher box office returns.

IMDB Rating vs Revenue



DATA ANALYSIS III: Runtime vs Revenue

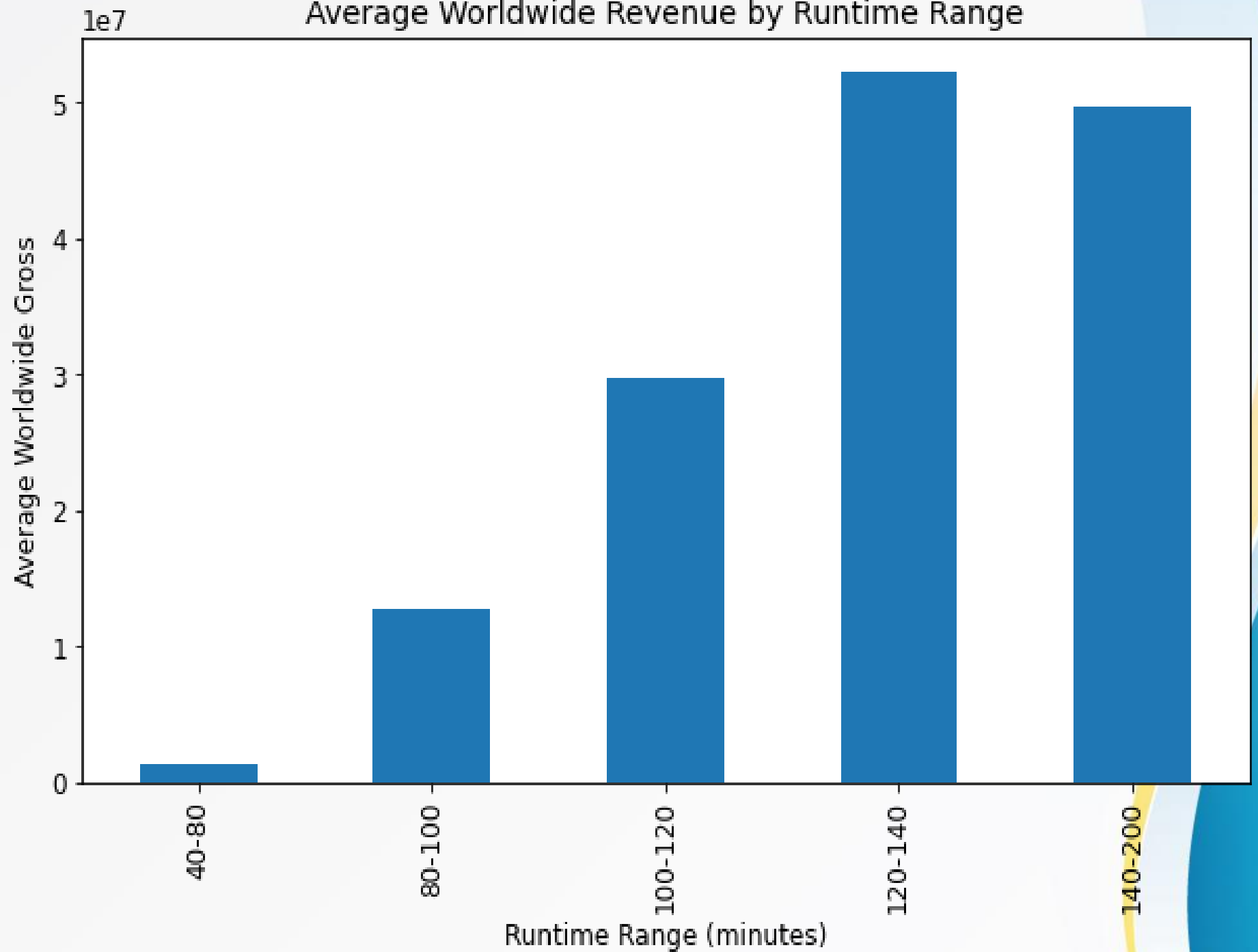
Findings:

- Films between 120–140 minutes earn higher revenue
- Very short films underperform
- Longer films can succeed if well executed

Recommendation:

- Audiences prefer feature-length films

Average Worldwide Revenue by Runtime Range



BUSINESS RECOMMENDATIONS

1. Prioritize Genre:

- Focus development on top-tier global genres identified in the analysis.

2. Target Quality:

- Aim for a "sweet spot" of audience approval (IMDb 6.5–8.0) through excellence in casting and production.

3. Optimize Length:

- Target a runtime between 2 and 2.3 hours to align with the highest-earning historical data.

NEXT STEPS

- Analyze production budgets vs returns and release timing analysis
- Assess marketing spend impact
- Explore regional revenue trends to determine the most profitable.
- Test combined genre, budget, and runtime strategies

THANK YOU

- Thank you for your time.

Q&A

- We welcome your questions.
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